Evaluating AI Agents on a Head-Gasket Repair Procedure

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Abstract

Can an AI agent carry out a mechanical repair, rather than only describe one? We present a head-gasket replacement benchmark built on a detailed CAD model of a 1990s 1.6-litre SOHC inline-four engine. Working through tool calls, the agent must look up the service specifications, dismantle the engine in an order the geometry allows, measure the head, block and bolts, choose the correct parts, tighten the head bolts in the specified stages and sequence, and reassemble the engine. We adopt TheAgentCompany's full and partial completion formulas without modification; our checkpoints are binary and checked by deterministic code. As a domain-specific cost, every action is also charged an estimated technician time, so mistakes and repeated work carry a labour cost. In exploratory runs, one per model on the standard job, no agent completed the repair fully correctly. GPT-6-Astra reached the highest partial score, 41.7% (15 of 18 checkpoint points), with all ten head bolts tightened correctly and 1,404 seconds of penalty time. GPT-6-Luna scored 36.1%. GPT-6.1-Sol scored 8.3% and left the engine disassembled. The most common failures were repeated work, repeating requests the environment had refused, and finishing with parts not refitted. These are single runs; they describe behaviour and do not rank the models.

Model names are as requested from the provider at xhigh reasoning effort; the serving system did not attest the model identity.

1 Introduction

A correct repair depends on decisions made along the way. A bolt tightened to the final value can still have an incorrect stage history. A head cannot be removed through an obstruction merely because its final pose is legal. A plausible written procedure does not show that an agent can maintain these constraints while acting.

We study a head-gasket replacement on a real 1990s 1.6 L SOHC inline-four. The agent

works through tools against a detailed CAD-based engine model on a prepared service stand. The surrounding vehicle is outside the task. Specifications must be found in the service-information corpus rather than supplied in the task prompt. This benchmark follows Engine Bench (https://toolsenabled.ai/bench/, code at https://github.com/ToolsEnabled/engine-bench) and uses the same engine model: Engine Bench asks an agent to build the engine, and this benchmark asks it to repair one.

We follow TheAgentCompany's organization and aggregate scoring [1]. Our contribution is the repair application: an environment where part identities, access, measurements, tightening stages and assembly completeness matter jointly. We do not propose a new scoring formula or statistical method. Technician time adds a domain-specific cost alongside computational cost.

The report distinguishes the task specification from supported software behavior and physical qualifications. Current results describe exploratory standard-job trajectories; the larger comparative collection remains pending. Historical scripted references and model attempts retain their actual rules and evaluator versions.

2 Benchmark desiderata and comparison to other benchmarks

The benchmark should require consequential actions, preserve intermediate state, and provide an inspectable basis for every score. An agent should be able to discover the information needed to act, and the evaluator should distinguish an incorrect repair from an unavailable measurement or an apparatus failure.

TheAgentCompany evaluates professional work through workplace tools and simulated colleague conversations, combining full completion with intermediate checkpoints. We adapt its reporting structure to a component-level repair procedure. The contribution is this application and its task specification, rather than a new evaluation method [1].

ScienceWorld links scientific knowledge to actions in a text simulation. SimulCost studies accuracy and computation cost in solver tuning. CompMat-Bench grades materials-research inputs and outputs, while its longer workflows also execute calculations. ReFract tests action trajectories against role-specific boundaries. DeepTest tests automotive information retrieval for omitted warnings. These settings illuminate different aspects of procedural work; their scores measure different targets and are not directly comparable [2] [11] [12] [13] [14].

Evaluation practice also shapes the design. AI Agents That Matter motivates cost-controlled comparisons, suitable holdouts and reproducible evaluation. The Agentic Benchmark Checklist distinguishes task validity from outcome validity. Harness-disclosure work makes the execution configuration part of the experimental condition, while the double-measurement study shows why evaluator truth and scaffold ownership require separate scrutiny. These sources motivate explicit task predicates, isolation, configuration disclosure and accounting for unavailable observations [3] [4] [5] [6].

Our desiderata are a documented service procedure, separate component identities, geometry-based access, stateful tool actions, task-specific deterministic checks, and reproducible initial conditions. Difficulty should arise from the job rather than hidden specifications or arbitrary tool failures. Missing service limits and unresolved access must remain visible.

3 Environment

The game presents a prepared engine, tools, replacement supplies and service information. The interface supports looking, inspection, measurement, service search, removal, installation, cleaning, replacement, tightening, timing checks and final submission. Every action refers to a specific target; attempts and state changes are recorded separately.

CAD bodies retain individual identities. A service-body index links procedure names to

geometry, but a display or index entry alone does not prove that a part can be withdrawn. Geometry and access checks must determine permitted removal order. Unknown constraints cannot be interpreted as permission.

Clamp mechanics provide state changes and observations such as bolt stretch, gasket condition and local clamp response. Unloaded service flatness is supplied or measured state; loaded head bow and bolt stretch are model responses. Available models include idealized material and contact behavior. These models support environment realism within their stated scope; they are not a full engine simulation or a validation of service life.

Neither a soak nor an engine run is part of scoring. The earlier operating-soak approach was dropped because the intended outcome is completion of the repair procedure against a correct reference. This is a design decision, not an engine-durability result.

4 Task structure

Each task comprises an instruction, a fixed starting state, a service corpus, weighted checkpoints and a deterministic evaluator. The proposed evaluator will read the final state and chronological action record. Binary checkpoint decisions will expose the supporting observations and the reason for each failed check.

The agent is asked to complete the repair, not to reproduce an answer string. The correct scripted baseline supplies the reference procedure and must satisfy every checkpoint. It defines 100% reference accuracy; that requirement is distinct from the complete original-rule references reported separately and from validation under the amended rules.

4.1 Evaluation metrics

Following TheAgentCompany, full completion is $S_{full} = 1$ if every checkpoint passes, and 0 otherwise. Partial completion is $S_{partial} = 0.5 \times (\text{points earned} / \text{total points}) + 0.5 \times$

S_full [1].

The aggregate formula is used unchanged. This application uses binary deterministic checkpoints. The template also permits partial credit within checkpoints and LLM evaluation for some tasks; those mechanisms are not used for the proposed repair checks [1].

Success is mean S_full and Score is mean S_partial. Steps counts model calls, not tool calls. API-equivalent Cost is prompt tokens multiplied by the prompt-token rate plus completion tokens multiplied by the completion-token rate, under the template's no-caching convention. Token counts, rate date, model identity and currency must be reported. A subscription fee is not an API price; unavailable rates or model-call counts remain unavailable [1].

Technician time is the sum of cited action charges and mistake-correction charges, reported as productive and penalty components beside the scripted baseline. It is charged time, not a measured mechanic's elapsed time or a simulated delay. Mistakes by type, agent wall time and tool calls are additional measures.

Secondary measures include each of the ten head bolts' condition, both torque stages and sequence position, plus checkpoint pass rates and failure categories. The explicit head-bolt group defines the denominator; covers and other fasteners do not enter it.

4.2 Workflow

Initialization restores the declared starting state, selects the variant, verifies the environment and evaluator versions, and starts the isolated collector. The task prompt identifies the job; the service corpus supplies specifications. Execution records each tool request, response, state transition and charge. Finalization evaluates the repair and preserves the complete report.

An end report is required after normal submission, budget exhaustion, early stopping, interruption or an apparatus refusal. It retains observed fields and named unknowns. Calling

finish does not by itself establish complete or correct work. Earlier mistakes remain in the action record; the checkpoint specification must state how later corrections affect final accuracy and charged corrective work.

Environment and workflow overview

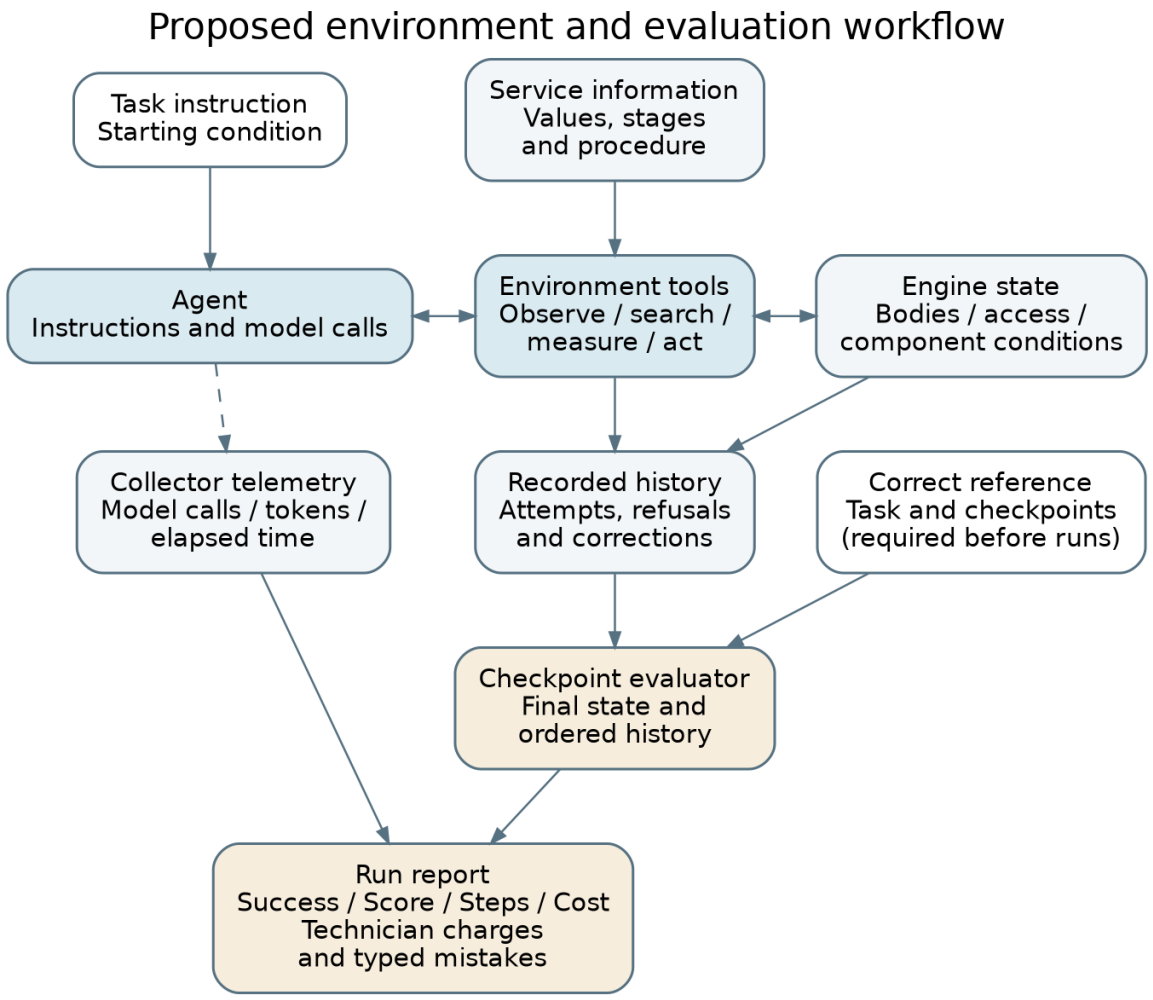


Figure 1. Proposed evaluation workflow. Tool observations and actions connect the agent to the engine state; the evaluator uses component state and ordered history. Dashed flow denotes usage telemetry. A complete reference procedure and collector remain requirements for study runs.

5 Task creation

The draft task set varies the condition of the same repair problem rather than introducing

unrelated jobs. It includes a standard job, a stretched bolt, a warped head, an incorrect gasket among available parts, and a worn timing belt. Each variant needs a realizable starting state, appropriate observations, a correct baseline and a tested evaluator before use.

Checkpoint curation follows the service procedure. It covers lookup, disassembly, timing preparation, loosening, cleaning, measurement, part choice, tightening and reassembly. Point values express task importance. They must be reviewed and frozen before collecting comparative runs; they must not be tuned to a model ranking.

The historical per-operation labor table consists of analyst estimates and charges successful actions only. Later scripted reports include corrective penalties, also as analyst estimates. Neither establishes the cited professional action-time and correction schedule required by the new design. Labor sources, charge units and rules for blocked attempts and rework remain to be established.

The earlier snapshot did not demonstrate the full task set. Later original-rule records contain complete references for every configured primary variant, including a software head-machining route and a separate head-replacement alternative. These records do not establish physical machining, exact service applicability or completion under the amended rules. Selected component transitions must remain bound to their actual runtime and configuration.

Each task should specify the starting component condition, the information visible to the agent, permitted service actions and the state or history evidence needed for every checkpoint. For each proposed variant, a reference procedure should demonstrate that the task is feasible under the same tool contract. Review should cover correct completion, plausible incorrect completion and unavailable measurements before model collection. The current task variants are design requirements; legacy fixture names do not establish their implementation.

6 Baseline agents

The planned agent is Codex CLI with only the environment's MCP tools, run through an isolated collector. The planned model arms are GPT-6.1-Sol, GPT-6-Astra and GPT-6-Luna. A Claude arm is a possible later extension.

The scripted correct repair is the environment reference, not a competing language model. It must execute the intended job with valid access and actions and satisfy the complete checkpoint set. Deliberately incorrect jobs test evaluator sensitivity to wrong order, wrong torque, omitted stages, incorrect parts, missed measurements and timing errors.

The exploratory run budget and earlier attempts remain in the accounting. Actual standard-task observations are reported below; the larger model-by-variant matrix remains pending. Requested and served model identifiers, transport refusals and stopping states retain their recorded meanings.

Report requested and returned model identifiers separately, with agent version, prompt, decoding controls, budgets, tools, context handling and retry policy. A configured model name does not prove the served model. Tool transport or process-cleanup fixtures do not establish production collection behavior [5].

An agent condition consists of the model and its execution harness. The collection record should specify the declared model version, inference settings, instructions, context policy, tools, stopping rule and retries. Conditions should receive the same task information and evaluation contract. Complete original-rule references are separate from the earlier incomplete diagnostics. References under the amended rules still need their own validation. Scripted usage fields and unavailable model telemetry must retain their recorded meanings [5].

7 Experimental results

Table 1. Experimental results

Agent	Model	Success	Score	Points (of 18)	Steps	Cost	Technician time, s (productive + penalty)	Mistakes
Scripted reference	n/a	100%	100%	18	n/a	n/a	13,630 + 0	0
Codex CLI	GPT-6-Astra	0%	41.7%	15	n/a*	n/a*	16,190 + 1,404	30
Codex CLI	GPT-6-Luna	0%	36.1%	13	n/a*	n/a*	15,230 + 14,970	309
Codex CLI	GPT-6.1-Sol	0%	8.3%	3	n/a*	n/a*	13,280 + 12,012	245

* The collector did not expose model-call counts, and no API-equivalent prices were applied; tool calls and token counts are in Appendix F.

Table 1 reports one exploratory run per model on the standard job (V1), with the scripted reference for comparison. No run achieved full completion. GPT-6-Astra finished the job in 528 tool calls and lost three points: two for attempting to remove a timing-cover screw that was still held (CP02), and one for an extra head-bolt loosening pass (CP04). Its technician time was 17,594 seconds, of which 1,404 were penalties, against 13,630 seconds for the reference. GPT-6-Luna also finished, but its penalties (14,970 seconds) nearly equalled its productive time. It lost points for blocked removals, a missed cleaning step before the final gasket placement, and fasteners and connectors not refitted. GPT-6.1-Sol did not finish: after using its only replacement gasket and later removing it, it could not complete the repair and left the head, gasket and bolts off the engine. Under the current version of the independent scorer, the two scorers agree on every checkpoint and every time charge for all three runs. An earlier version differed by 30 seconds on GPT-6.1-Sol because it did not classify one gasket refit; this was corrected (Appendix F). With one run per model, these figures describe individual runs; they are not estimates of model performance.

7.1 Common failures

(a) Repeated and excess work

Repeated work accounted for 213 operations and 10,350 penalty seconds for GPT-6.1-Sol, 15 operations and 630 seconds for GPT-6-Astra, and 191 operations and 9,840 seconds for GPT-6-Luna. Sol and Astra each loosened the ten head bolts in four passes instead of three. Their extra passes—Sol 220–229 and Astra 240–249—each added 420 seconds and lost CP04's one point. Luna removed the head and fitted the gasket again at 692–693; both operations were charged as repeated work.

(b) Repeating refused requests without changing the repair

GPT-6-Luna made 40 consecutive restoration requests at 424–463, receiving the same refusal every time: "Restore attachments after all selected service positions are installed". The requests targeted different fasteners, but no parts were refitted between them to meet that requirement. This block cost 1,200 seconds. All 43 restoration refusals were included in Luna's 110 invalid actions, charged 4,890 seconds overall. The refusal named the unmet requirement but not the missing parts; the next environment version adds the requested item and the number of parts not yet refitted.

(c) Parts and consumables

GPT-6.1-Sol discarded the old gasket at 276, then fitted its only new gasket at 327 by installing it directly from stock rather than through the replace action, which records a new part as the replacement for the original. Later replace requests at 438 and 661 both returned "Replacement supplies unavailable". The procedure does not refit a multi-layer steel gasket once removed, so removing it at 660 left Sol with no usable gasket.

(d) Order-of-work requirements

GPT-6-Astra released a timing-cover screw at 112, then tried to remove it at 113 while the crank pulley was still in place, blocking the screw's withdrawal route. The refusal cost 30

seconds and CP02's two points. Sol's three blocked removals and Luna's seven similarly cost 90 and 210 seconds, with CP02 unmet in both runs. Separately, Luna removed the head at 692 and placed the final gasket at 693 without cleaning in between. Earlier cleaning at 355–358 did not meet this requirement; CP05 lost one point.

(e) Finishing with work outstanding

GPT-6-Luna submitted finish at 807 with 100 fastener and connector restorations still outstanding. It failed the two-point final-reassembly check, CP12, and received a further 30-second missing-restoration charge. GPT-6.1-Sol never submitted finish: after 659–660, the head, gasket and all ten head bolts remained in the tray. Its final state failed the five two-point completion checks, CP08–CP12, despite earlier successful tightening at 339–358.

The final scores were 3/18 for GPT-6.1-Sol, 15/18 for GPT-6-Astra and 13/18 for GPT-6-Luna. Penalty totals were 12,012, 1,404 and 14,970 seconds respectively, agreed by the current version of both scorers. The earlier independent scorer recorded 11,952 penalty seconds for Sol (Appendix F). These examples show how repeated work, unmet prerequisites and incomplete reassembly accumulated costs even when individual repair operations succeeded.

8 Implications and future directions

A repair environment can test whether an agent uses information while maintaining a changing physical-state model. The application extends a professional-task benchmark pattern into mechanical procedure execution through software tools. Its value depends on accurate task constraints and evaluators, not on adding a new aggregate score. Engine Bench and this benchmark share one engine model and together cover building the engine and repairing it; both are published at https://toolsenabled.ai/bench/, with code at

https://github.com/ToolsEnabled/engine-bench.

Future work should complete the service-to-geometry mapping, validate references under the amended rules, validate the technician-time estimates against published labour-time guides and extend the original-rule software checks to the selected collection runtime. Collected trajectories can then support analysis of procedural decisions and resource use. Broader tasks and physical execution require separate evidence.

The benchmark does not certify that an agent or robot can safely repair an engine. It omits the surrounding vehicle, workshop dexterity and many material uncertainties. ToolsEnabled, Inc. develops the environment and authors this report; this relationship should be considered when interpreting the design. The exploratory attempts do not establish comparative population performance.

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Appendix A. Benchmark comparison

Selected task environments and evaluation targets. This descriptive comparison adapts the benchmark-comparison organization of TheAgentCompany to a narrower repair application; it is not a reproduction of its feature table or a ranking [1].

Table 2. Appendix A. Benchmark comparison

Benchmark	Task and interaction	Evaluation target	Distinction for this application
TheAgentCompany	Professional tasks across workplace tools, with simulated colleague conversations	Full completion and weighted intermediate checkpoints; evaluators include programmatic checks and language-model judgments	Provides the reporting structure; colleague interaction differs from repair-tool interaction
ScienceWorld	Elementary science procedures in a text simulation	Required goals and optional progress subgoals, with task variations	Connects knowledge to state-changing actions; spatial and physical models are deliberately simplified
SimulCost	Parameter selection and iterative tuning through physics solvers	Accuracy thresholds and simulation cost relative to a scan reference	Separates tool computation cost from inference cost; technician time is a different application quantity
CompMat-Bench	Materials-research input preparation, output analysis and calculation workflows	Fixed checks against reproduced research results; single tasks use precomputed expensive simulations	Workflow agents also run calculations; the target is research reproduction rather than equipment repair
ReFract	Role-conditioned action trajectories in executable text world models	Goal attainment, role-boundary compliance and call efficiency	Tests declared capability and knowledge boundaries in a closed simulation; it does not establish physical equipment consequences
DeepTest	Generated requests to automotive manual-retrieval assistants	Warning omissions, failure-revealing test coverage and an LLM-based oracle	Evaluates testing tools and information reliability rather than executed repair procedures
Proposed repair application	Component inspection, part selection and ordered service actions through an environment interface	Binary weighted checkpoints, full completion, inference cost and charged technician time	Application design with unresolved service and evaluator details; comparative model results are pending

Sources: [1] [2] [11] [12] [13] [14]. Selected scopes do not imply equal difficulty. Colleague conversation, physical-model computation and repair-tool actions are distinct interactions.

Appendix B. Environment and engine model

The service-body index contains 1240 selected body rows. Individual bodies remain distinct, including hardware candidates and contextual geometry. The mapping provides a basis for task-specific access checks; it is not proof of a complete legal disassembly route.

Table 3. Appendix B. Environment and engine model

Tool group	Role in the repair
look / inspect / measure	Expose visible state and permitted component measurements.
search_service_info	Retrieve specifications from the task corpus.
remove / install	Change component placement under access and dependency constraints.
clean / replace	Change preparation state or exchange an identified component.
torque / check_timing	Record ordered tightening and timing checks.
finish	Submit the final state; revised evaluator integration remains to be demonstrated.

The inherited interface includes a historical operating-harness endpoint. Its action signatures can support the revised design, but its previous finish behavior is not evidence of the new checkpoint evaluator.

The service corpus distinguishes published service information, private source references, benchmark procedure choices and estimates. Selected torque transcriptions differ, reuse limits are incomplete, and gasket orientation and dowel fit need explicit support. Those gaps must be resolved or declared before a task can claim a determinate correct answer.

The environment exposes component observations and state-changing actions through an interface. Existing software demonstrations support component identity and selected action behavior. The engine display supplies geometric context, while the mechanical models make finite, assumption-dependent predictions. A complete display does not establish a usable service path, and a model response does not validate the service procedure. Material history, unloaded flatness and loaded deformation must be distinguished when defining measurements.

Appendix B. Benchmark construction

Construction links the service procedure to observable component state and permitted actions. The following review stages separate task feasibility from evaluator behavior; none substitutes for a completed reference run.

Table 4. Appendix B. Benchmark construction

Construction stage	Evidence and review target
Define the job	Starting component conditions, supplied tools and parts, visible information, terminal state and legitimate waste dispositions.
Map service information	Applicable specifications, measurement units, tightening stages and sequence, reuse and replacement decisions; unresolved limits remain explicit.
Map components and actions	Stable body and fastener identities, access dependencies, state changes, refused actions and observations available to the agent.
Curate checkpoint evidence	Final-state and history predicates, weights, unavailable evidence, correction rules and failure explanations.
Exercise reference and faulty jobs	Feasible correct execution plus controlled departures that test evaluator sensitivity; keep software diagnostics distinct from model performance.
Freeze collection configuration	Task, service corpus, geometry, evaluator, agent harness, budgets, telemetry and charge schedule fixed before comparative collection.

Reviewers should inspect a complete example from instruction through final report. Changing a service threshold, active component identity or evaluator condition changes the task definition and requires a new reviewed configuration. Historical software tests remain tied to the versions they exercised.

The factory-style path lifts a mapped rocker assembly of 109 bodies with 15 bolts retained in that group. It requires 14 releases of head threads and 16 adjuster backoffs; the camshaft remains installed and needs independent support. A full-strip path remains a valid alternate. Group retention, support and unobstructed withdrawal retain provisional geometric qualifications.

For the default fixture, the grouped route has 162 removal operations and 51 moving groups, compared with 185 operations and 71 groups for the full-strip route. Both restore 626 removed positions and retain 614 bodies. Alternative stay choices carry their own checklists. These are removal-fixture counts, not complete repair action counts or technician seconds; neither path is physical service validation.

Appendix B. Engine geometry

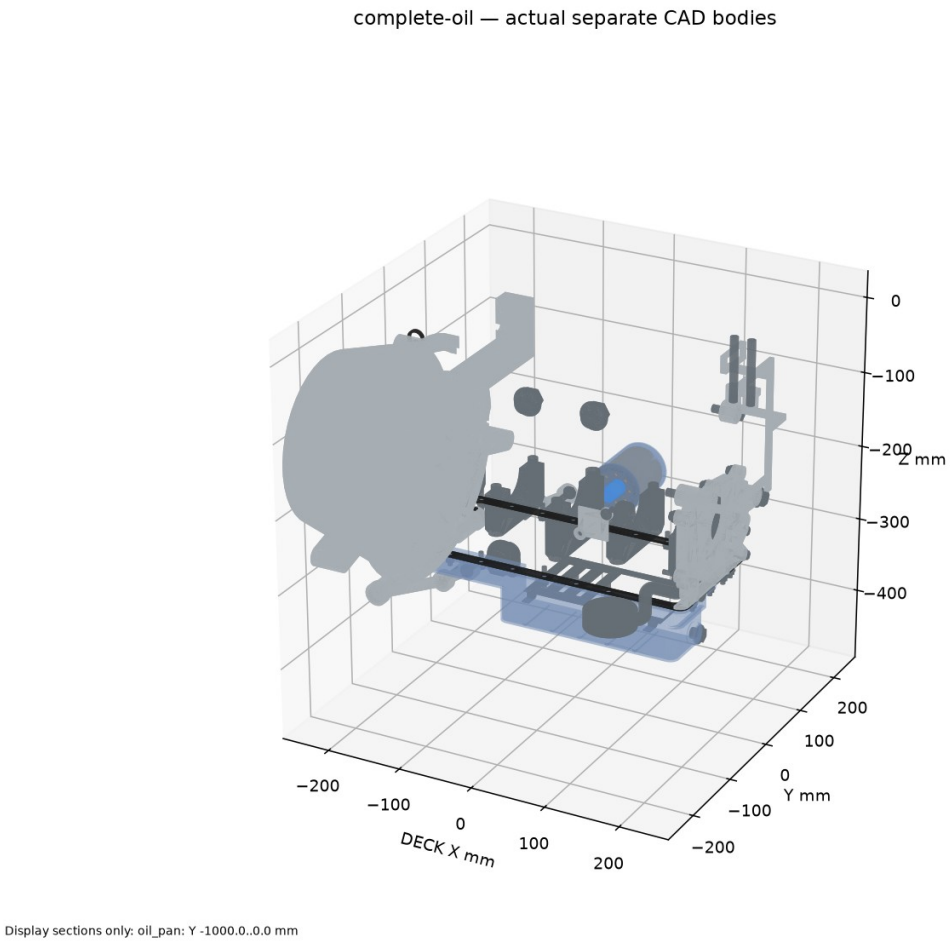


Figure 2. Lubrication component representation. Separate bodies and their context are preserved. This view does not establish an unobstructed service path. In the source render the vertical-axis label overlaps one tick label; the illustration is reproduced without alteration.

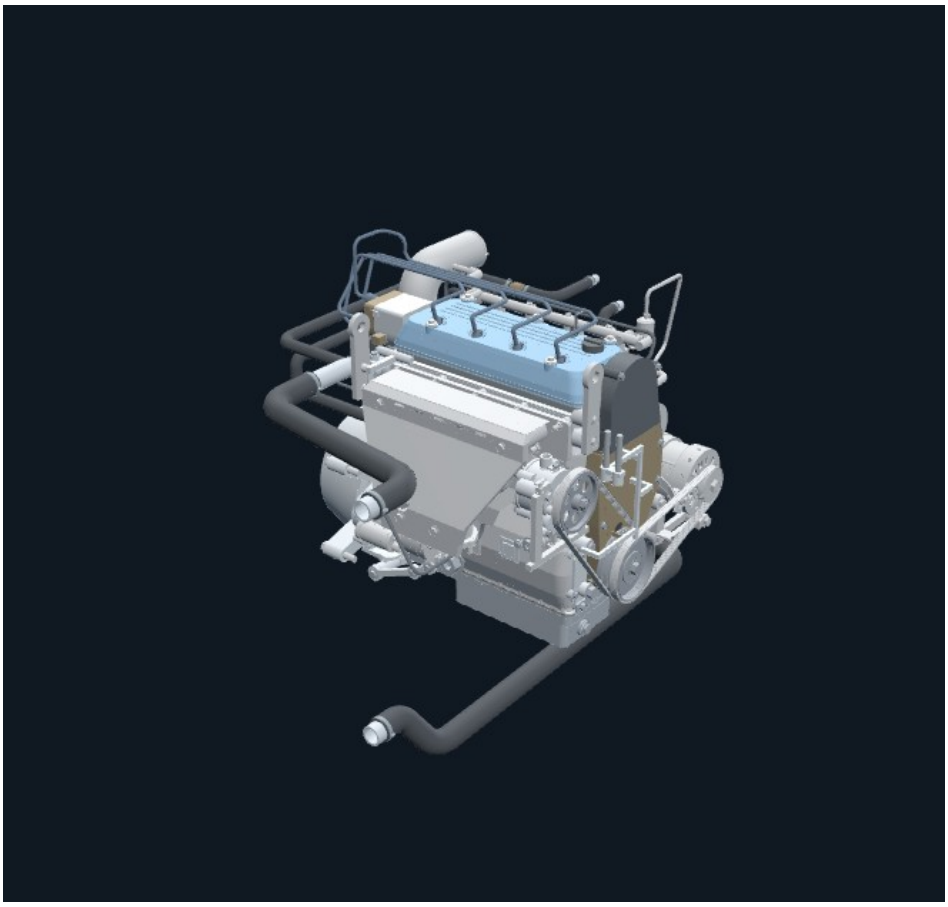


Figure 3. Integrated engine display. The original component geometry is shown without changing its body identities. Display completeness does not resolve all access and intersection questions.

Appendix B. Mechanical assumptions

The mechanical model links bolt loading, gasket compression and head response. The historical rigid model and later elastic plate approximation have different scopes. A linear elastic mechanical plate with a yield exclusion must not be described as a model of permanent head deformation.

Separate generic material-law work carries gasket recovery, bolt stretch and sectional material history. Its parameters and applicability are not selected-part validation. Finite order comparisons show that an action history can affect transient component response; they do not establish lasting damage, reliable repair classification or engine durability.

Material proxies, contact reduction and finite comparison limits remain explicit. Mechanical state is used to make inspection and measurement meaningful. Checkpoint correctness is determined by the reference procedure and recorded work, without operating the engine.

Appendix B. Finite component illustration

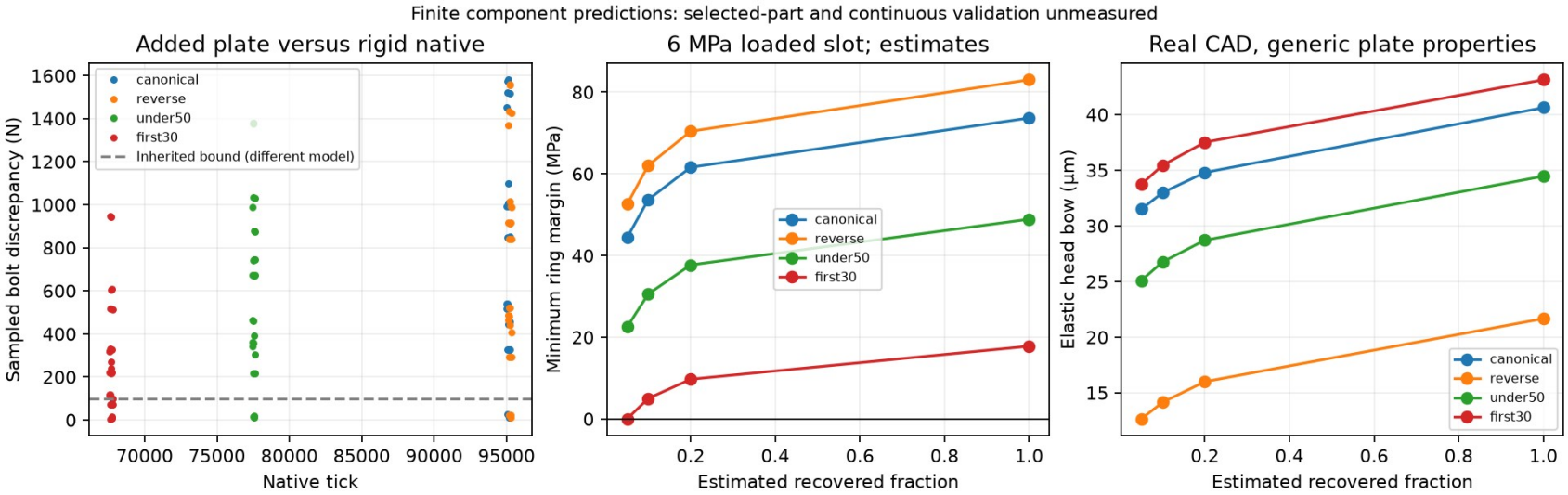


Figure 4. Finite component-model predictions retained as an illustration of mechanical assumptions. The dashed inherited-model reference does not validate the added plate model. Recovery and material properties are estimates. Positive margins for incorrect command sequences are not evidence of a correct repair.

Appendix C. Task and checkpoint details

These variants are a draft task set. All checkpoints retain their normal meaning when an abnormal condition is absent. Variant identifiers are task labels, not observed samples.

Table 5. Appendix C. Task and checkpoint details

Variant	Required decision
V1 standard job.	Nothing abnormal.
V2 stretched bolt.	One head bolt is beyond the reuse limit and must be found by measurement and replaced.
V3 warped head.	The head is beyond the flatness limit and must be measured and sent for machining, or replaced, before reassembly.
V4 wrong gasket offered.	The parts available include an incorrect gasket next to the correct one.
V5 worn timing belt.	The timing belt must be inspected and replaced.

Original-rule references cover the warped-head and worn-belt configurations, with a separate head-replacement alternative. This supports the recorded software cases rather than physical machining or part-fit validation. Component identity, condition and timing transitions must be reviewed against the exact amended runtime before comparative collection.

A proposed wrong-gasket task offers a specified incorrect gasket alongside the correct replacement. The agent must inspect the assembly, identify the required replacement and complete the applicable service sequence. A final declaration alone cannot demonstrate part identity or ordered torque stages; checkpoint evidence must come from component state and action history. Reassembly checks should distinguish legitimate replacement and disposal from an absent required component. This example illustrates the proposed evaluation contract and is not an observed agent run.

Appendix C. Signed checkpoint requirements

Table 6. Appendix C. Signed checkpoint requirements

Checkpoint	Required evidence	Points
1	Torque values, both stages and the sequence were retrieved from the service information before the first tightening of any head bolt.	1
2	Disassembly followed the documented removal policy. Any refused removal attempt of a part that was still blocked or still attached fails CP02, and is also counted as a mistake.	2
3	Before the timing belt came off: cylinder 1 at TDC and the timing marks aligned, confirmed by a check observation.	1
4	Head bolts loosened in the specified order and stages before head removal. The agent must use one complete documented sequence, either the factory diagram order (10, 9, 8, 7, 6, 3, 4, 5, 2, 1) or exact reverse (10 to 1), consistently across all loosening passes.	1
5	After the head and the old gasket were removed, and before the new gasket was placed: both mating surfaces cleaned.	1
6	Head and block flatness both measured and compared with the service limit before the head was installed. The decision is correct for the variant: V3 machine or replace the head; otherwise reuse it.	1

Appendix C. Signed checkpoint requirements continued

Table 7. Appendix C. Signed checkpoint requirements continued

Checkpoint	Required evidence	Points
7	Every head bolt measured against the reuse limit before its keep-or-replace decision, and the decision is correct: V2 replace the stretched bolt; otherwise keep.	1
8	A new, correct head gasket (75 mm bore, 0.762 mm MLS) installed in the correct orientation with the dowels aligned. V4: installing the 78 mm / 1.2 mm part fails.	2
9	Final state: all ten bolts carry a valid stage-1 torque at the specified value, applied in the specified sequence.	2
10	Final state: all ten bolts carry a valid final-stage torque at the specified value, applied in the specified sequence after stage 1.	2
11	Timing restored: marks aligned and belt tensioned, confirmed by a successful check after the belt was reinstalled. V5: a new belt.	2
12	Reassembly complete: every removed part and every released fastener or attachment restored, or properly superseded by its replacement, per the documented restoration checklist. The sources make a new head gasket mandatory, but for the valve-cover gasket they describe inspection, reuse when serviceable, and sealant treatment. CP12 therefore requires a new head gasket (unchanged), and either a new valve-cover gasket or the original inspected and refitted with sealant as the manual describes. The earlier "both new" rule was a benchmark policy, not a manual requirement, and is withdrawn.	2

Total available points: 18. Each checkpoint is binary, and all checkpoints remain applicable to every task. A correct normal-condition decision can pass a measurement checkpoint without replacing the part.

The current amendments accept either documented loosening sequence, used consistently across all loosening passes, and withdraw the unconditional new cover-gasket policy. The new head gasket remains mandatory. Reassembly restores required components with legitimate replacement and waste dispositions. Final-state corrections may recover credit; chronological process violations and charges remain recorded. Historical commands and reports retain their original source versions.

Appendix C. Engineering specifications — H01 to H05

The complete specification table distinguishes public-source support from benchmark policy. The captured public workshop manual identifies its first edition as 9/91. Exact reconstructed-hardware applicability and later manual supplements remain unverified. Policy rows make no factory-value claim. Every source disagreement and row scope is retained below. The table records the earlier checkpoint amendment; the current requirement to use one chosen loosening sequence across all passes appears in the checkpoint table above. Full original fields, source URLs and implementation locators accompany the report.

Table 8. Appendix C. Engineering specifications — H01 to H05

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
H01: First complete torque pass	30 N m; active benchmark	public source-supported; M|S2; 30; M 6-54; S2 torque steps	Selected two-stage related-application procedure
H02: Final complete torque pass	73 N m; active benchmark	public source-supported; M|S2; 73; M 6-54; S2 torque steps	Retain matching workshop and fastener-sheet value S1 related article converts final torque to72 Nm; other engine variants use different procedures.
H03: Tightening stages	2 complete passes; active benchmark	public source-supported; M|S2; 30 then73; M 6-54; S2 steps	Each pass completes all ten positions before the next This is installation stage2, not later thermal-service retorque.
H04: Number of service positions	10 bolts; active benchmark	public source-supported; M|S2; 10; Head torque diagram	Selected service layout
H05: Tightening position order	1,2,3,4,5,6,7,8,9,10 position sequence; active benchmark	public source-supported; S2; ascending numbered positions; Torque-order diagram	Diagram numbering controls sequence

Appendix C. Engineering specifications — H06 to H10

Table 9. Appendix C. Engineering specifications — H06 to H10

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
H06: Diagram row layout	upper:8,6,2,4,10; lower:9,3,1,5,7 positions; active benchmark	public source-supported; S2; same two rows; Torque-order diagram	Related diagram numbering Mapping diagram upper/left to runtime +Y/+X is benchmark coordinate convention; exact CAD landmark correspondence remains unverified.
H07: Retained scripted loosening order	10,9,8,7,6,5,4,3,2,1 position sequence; retained scripted reference; current acceptance listed in H24	benchmark policy choice; S1|M; S1 reverse instruction; M diagram differs at two lower-row positions; S1 head-removal prose; M 6-30 and6-54 diagrams	Preserve the exact recorded reference; Amendment 1 also admits the M diagram order M loosening upper:3,5,9,7,1; lower:2,6,10,8,4. This recorded reference is not the only accepted CP04 sequence.
H08: Loosening increment	-120 degrees/bolt/pass; active benchmark	public source-supported; M|S1; one-third turn each pass; M 6-30; S1 printed page16	Negative API angle denotes loosening Fixed number of passes follows software engagement, not physical thread length.
H09: Scripted initial engagement	1 turn; active benchmark	benchmark policy choice	Finite procedural disengagement state Not derived from140 mm nominal bolt length.
H10: Initial reference loosening passes	3 passes; active benchmark	benchmark policy choice	Three one-third-turn passes release the one-turn initial state

Appendix C. Engineering specifications — H11 to H15

Table 10. Appendix C. Engineering specifications — H11 to H15

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
H11: New installation engagement	2 turns; active benchmark	benchmark policy choice	Discrete started-fastener state in fixture path Source-defined initial_conditions can override; not measured engagement.
H12: Initial assembled torque	73 N m; active benchmark	benchmark policy choice; M|S2; 73; Related final tightening value	Authored initial state initialized to final service value
H13: Thread nominal diameter	10 mm; active benchmark	public source-supported; S2; 10; 704.700 bolt dimension	Related fastener dimension
H14: Thread pitch	1.25 mm; active benchmark	public source-supported; S2; 1.25; 704.700 bolt dimension	Related fastener dimension
H15: Nominal bolt dimension	140 mm; active benchmark	public source-supported; S2; 140; 704.700 bolt dimension	Related nominal part length Not a measured free length or allowable reuse limit.

Appendix C. Engineering specifications — H16 to H20

Table 11. Appendix C. Engineering specifications — H16 to H20

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
H16: Lubrication	threads and washer bearing surface oiled criterion; active benchmark	public source-supported; M|S1; engine oil on threads and bearing faces; Head installation	Service preparation condition
H17: Replacement threshold	0.1 mm extension; active benchmark	benchmark policy choice	Deliberate measurable V2 decision boundary; no applicable numeric factory reuse threshold established Replace strictly above threshold; equality does not trigger this rule. No failure-time inference.
H18: V2 supplied bolt extension	0.2 mm; active benchmark	benchmark policy choice	Authored fault exceeds0.1 mm benchmark threshold
H19: V2 service position	3 position; active benchmark	benchmark policy choice	Fixed variant instance makes the defect discoverable
H20: Other/new bolt extension	0 mm; active benchmark	benchmark policy choice	Authored unworn/replacement state Unknown measured free length must remain unknown.

Appendix C. Engineering specifications — H21 to C02

Table 12. Appendix C. Engineering specifications — H21 to C02

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
H21: Under-torque control final target	50 N m; active benchmark	benchmark policy choice	Scripted torque-value error relative to73 Nm
H22: Wrong-order control	reverse both passes criterion; active benchmark	benchmark policy choice	Tests procedural ordering independent of force uniformity
H23: Skipped-stage control	omit complete73Nm pass criterion; active benchmark	benchmark policy choice	Tests missing final installation pass
C01: Final torque	10 N m; active benchmark	public source-supported; M|S1; 10; M 6-56 cylinder-head-cover arrow; S1 printed page52 Valve Cover	Related service torque
C02: Intermediate torque passes	3,6 N m; active benchmark	benchmark policy choice	Scripted gradual seating; no corresponding manual numerical steps established

Appendix C. Engineering specifications — C03 to C07

Table 13. Appendix C. Engineering specifications — C03 to C07

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
C03: Service position count	5 bolts; active benchmark	benchmark policy choice	Selected software service layout Original cover pattern not established by the cited prose.
C04: Tightening order	1,2,3,4,5 position sequence; active benchmark	benchmark policy choice	Deterministic selected layout
C05: Loosening order and steps	5,4,3,2,1;3 passes of-120degrees sequence; active benchmark	benchmark policy choice	Symmetric procedural release rule Head-bolt one-third-turn instruction does not establish this cover pattern.
C06: Thread diameter/pitch	6 / 1 mm; active benchmark	public source-supported; M; M 6x1.0; M 6-56 cylinder-head-cover fastener arrow	Related cover fastener size
C07: Nominal length	40 mm; active benchmark	benchmark policy choice	Fixture fastener dimension

Appendix C. Engineering specifications — C08 to S02

Table 14. Appendix C. Engineering specifications — C08 to S02

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
C08: Initial assembled torque	10 N m; active benchmark	benchmark policy choice; M; 10; Cover installation	Authored initial state uses final service value
C09: Local oil and sealant treatment	oil tube seals; inspect for damage; sealant at specified cam-holder corners criterion; active benchmark	public source-supported; M|S1; local oil, damage inspection and liquid-gasket treatment; M 6-55 step12 and application note; S1 printed page17	Separate cover treatment from dry head MLS gasket Reviewed instruction does not establish unconditional new cover-gasket replacement.
C10: Retained reference replacement	new distinct cover gasket criterion; retained scripted reference; current acceptance listed in C11	benchmark policy choice	Preserve the replacement performed by the recorded reference The original mandatory both-new checkpoint rule is withdrawn. Do not cite the mandatory new HEAD gasket instruction as cover-gasket evidence.
S01: Head flatness decision limit	0.05 mm; active benchmark	benchmark policy choice; M; 0.05 mm conditional resurfacing criterion; Printed6-45	Simplify related conditional service guidance into one benchmark decision Real resurfacing also depends on head height and cam-holder clearances.
S02: Block flatness decision limit	0.05 mm; active benchmark	benchmark policy choice; M; 0.07 mm standard;0.10 mm service; Printed7-12	Common stricter benchmark limit across both surfaces Do not label0.05 mm as the public block service limit; S1 lists0.079 mm standard.

Appendix C. Engineering specifications — S03 to S07

Table 15. Appendix C. Engineering specifications — S03 to S07

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
S03: Normal initial head/deck/cover flatness	0.02 mm; active benchmark	benchmark policy choice	Authored standard-state measurements
S04: V3 initial head flatness	0.08 mm; active benchmark	benchmark policy choice	Authored head exceeds benchmark0.05 mm limit
S05: Material removed by machine_head	0.06 mm; active benchmark	benchmark policy choice	Deterministic service-result stand-in Eligibility does not model actual head height or cam-holder geometry.
S06: Post-service head flatness	0.02 mm; active benchmark	benchmark policy choice	Authored recovered state for machining or replacement
S07: Normal/post-service roughness	0.5 um Ra; active benchmark	benchmark policy choice; S3; 50RA or finer; page does not state a unit; Surface Finishing	Declared surface state, not a conversion of the webpage number

Appendix C. Engineering specifications — S08 to G01

Table 16. Appendix C. Engineering specifications — S08 to G01

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
S08: Abrasive-disc damage roughness	5 um Ra; active benchmark	benchmark policy choice	Authored adverse cleaning outcome No calibrated tool-to-damage law.
S09: Ready surface	clean; dry; undamaged; no debris criterion; active benchmark	public source-supported; S3|S4; clean dry mating faces; avoid damaging preparation; MLS installation; surface preparation	Represent inspection conditions separately from loaded deformation
S10: Unprotected cleaning debris increment	1 count; active benchmark	benchmark policy choice	Discrete penalty-state observation when openings are unprotected Not a measured particle-generation rate.
S11: After-clean residue	0 fraction; active benchmark	benchmark policy choice	Successful cleaning state transition No cleaning kinetics simulated.
G01: Correct selected bore	75 mm; active benchmark	benchmark policy choice; S6; 75.5 mm; C4251-030 catalog	Study-design selected stock; B01 overrides related catalog The catalog does not support75 mm as its part dimension.

Appendix C. Engineering specifications — G02 to G06

Table 17. Appendix C. Engineering specifications — G02 to G06

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
G02: Correct selected compressed thickness	0.762 mm; active benchmark	benchmark policy choice; S6; .030in = .762 mm; C4251-030 catalog	Selected stock definition agrees with related catalog thickness only
G03: Correct selected construction	multilayer steel material type; active benchmark	benchmark policy choice; S6; MLS;3layers; C4251-030 catalog	Study stock category; no selected measured constitutive curve
G04: Wrong offered bore	78 mm; active benchmark	benchmark policy choice	Deliberately different V4 stock identifier/dimension Not the historical2.5 mm offset profile.
G05: Wrong offered compressed thickness	1.2 mm; active benchmark	benchmark policy choice	Deliberately different V4 stock definition
G06: Wrong offered construction	multilayer steel material type; active benchmark	benchmark policy choice	Authored wrong-stock material category Dimensions do not establish a measured material curve.

Appendix C. Engineering specifications — G07 to G11

Table 18. Appendix C. Engineering specifications — G07 to G11

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
G07: Selected orientation	up enum; active benchmark	benchmark policy choice; M|S6; correct installation alignment; catalog None; Head installation; catalog Orientation field	Discrete orientation and dowel alignment convention Catalog None does not certify either-face installation.
G08: Sealant on selected MLS	none; install dry criterion; active benchmark	public source-supported; S3; dry coated MLS; no extra sealer; MLS Head Gasket Installation	Apply general MLS installation guidance
G09: Replacement	new head gasket criterion; active benchmark	public source-supported; M|S1; new gasket; Head installation	Used gasket is removed/replaced
G10: Dowel alignment	all retained dowels installed; selected orientation criterion; active benchmark	benchmark policy choice; M; alignment dowels; Head installation	Discrete benchmark alignment observation No clearance or geometric interference simulation.
G11: Failed-cylinder identifier	2 cylinder; active benchmark	benchmark policy choice	Fixed starting defect metadata across variants No combustion or leak-time simulation.

Appendix C. Engineering specifications — T01 to T05

Table 19. Appendix C. Engineering specifications — T01 to T05

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
T01: Compression TDC cylinder	1 cylinder; active benchmark	public source-supported; M|S1; No.1 compression TDC; M 6-62; S1p1	Selected service datum
T02: Compression-datum phase	cam0;crank0 degrees; active benchmark	benchmark policy choice	Explicit software coordinate origin Not a measured tooth/keyway angle.
T03: Cam/crank speed relation	cam=crank/2 ratio; active benchmark	benchmark policy choice	Discrete four-stroke phase convention No motion simulation in this benchmark path.
T04: Mark-alignment tolerance	0.1 degrees; active benchmark	benchmark policy choice	Finite exact evaluator tolerance Not a workshop measuring tolerance.
T05: Crank mark	tooth-side groove to oil-pump pointer alignment; active benchmark	public source-supported; M; same; Printed6-62	Selected related-application mark convention

Appendix C. Engineering specifications — T06 to T10

Table 20. Appendix C. Engineering specifications — T06 to T10

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
T06: Cam mark	rear-cover pointer; UP upward alignment; active benchmark	public source-supported; M; same; Printed6-62 selected application	Selected source application; other diagrams differ Head-surface/dual-pointer diagrams belong to other source applications.
T07: Reference seating/recheck turns	4 crank revolutions; active benchmark	benchmark policy choice; M; about 4 or 6 clockwise turns; Printed6-63	Keep existing scripted reference convention with correct source attribution S1 page20 says5–6 counterclockwise turns; conflicting direction/count explicitly retained.
T08: Reference seating direction	clockwise viewed from pulley side; active benchmark	benchmark policy choice; M|S1; M: clockwise; S1: counterclockwise; M 6-63 versus S1p20	Existing benchmark selects M for this seating step Documentation convention only: check_timing does not simulate or expose rotation direction. Tension adjustment is separately counterclockwise; E10 must not attribute clockwise motion to S1.
T09: Timing-check input domain	1 through6 integer crank revolutions; active benchmark	benchmark policy choice	API resource/domain bound Not proof that any permitted count completes a real service procedure.
T10: Pre-removal reference check	1 crank revolution; active benchmark	benchmark policy choice	Scripted compression-datum inspection before removal Discrete check; does not execute physical rotation.

Appendix C. Engineering specifications — T11 to T15

Table 21. Appendix C. Engineering specifications — T11 to T15

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
T11: Wrong-timing control phase	15 cam degrees; active benchmark	benchmark policy choice	Authored misalignment control at crank0
T12: Belt installation order	crank;adjuster;water pump;cam sequence; active benchmark	public source-supported; M|S1; same; M 6-62; S1p20	Related service sequence Abstract belt install is not a routed belt simulation.
T13: Tension-adjustment direction	counterclockwise pulley-side view; active benchmark	public source-supported; M|S1|A; same; M 6-58; S1p3; A PDFp6	Distinct from selected seating step Abstract tension_belt records state without these physical motions.
T14: Adjuster release for tension	180 degrees; related reference; abstracted operation	public source-supported; M|S1|A; 180; M 6-58; S1p3; A PDFp6	Related service instruction Reference instruction; abstract API does not expose adjusting-bolt motion.
T15: Tension-creating movement	3 cam-pulley teeth; related reference; abstracted operation	public source-supported; M|S1|A; 3 counterclockwise; M 6-58; S1p3; A PDFp6	Related service instruction Do not call this three crank revolutions.

Appendix C. Engineering specifications — T16 to W01

Table 22. Appendix C. Engineering specifications — T16 to W01

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
T16: Replacement criteria	cracks;oil/coolant contamination;tooth damage condition; active benchmark	public source-supported; S1; same; Printedp3	Inspect and replace damaged belt No numeric wear-life or maintenance interval modeled.
T17: V5 initial condition	damaged=true boolean; active benchmark	benchmark policy choice	Authored worn-belt task instance Boolean damage does not identify which physical inspection defect occurred.
T18: Replacement state	new distinct body;damaged=false criterion; active benchmark	benchmark policy choice	Preserve physical identity and corrected inspection state No extra stock timing mechanics inferred.
T19: Tensioned state	true after successful tension_belt boolean; active benchmark	benchmark policy choice	Explicit procedural operation state Not a measured belt force.
W01: Cold service temperature ceiling	38 degC; active benchmark	public source-supported; S1; less than38; Printedp3 timing adjustment	Related service precondition No thermal evolution simulated; do not generalize this to all unrelated tasks.

Appendix C. Engineering specifications — W02 to S15

Table 23. Appendix C. Engineering specifications — W02 to S15

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
W02: Isolation and support	battery isolated;coolant drained;engine supported criterion; active benchmark	public source-supported; M|S1; required preparation; Removal preparation	Service safety/preparation conditions Runtime preparation state remains discrete.
S12: Initial residue fraction	1 fraction; active benchmark	benchmark policy choice	Authored soiled starting condition
S13: Initial moisture and debris	dry=false;debris_count=0;damaged=false state; active benchmark	benchmark policy choice	Authored starting inspection state
S14: Conditional head material-removal maximum	0.2 mm; service corpus; not an implemented limit	public source-supported; M; 0.20 mm; subject to head height and cam-holder checks; Printed 6-45	Retain service-information context; not automatic permission to machine Benchmark machining eligibility does not perform those geometric checks.
S15: Block standard warpage maximum	0.07 mm; service corpus; benchmark uses0.05 mm	public source-supported; M; 0.07 mm; Printed 7-12	Correct public attribution for E06 S1 printed page46 gives0.079 mm; do not silently substitute its converted value.

Appendix C. Engineering specifications — S16 to G14

Table 24. Appendix C. Engineering specifications — S16 to G14

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
S16: Block service warpage limit	0.1 mm; service corpus; benchmark uses0.05 mm	public source-supported; M|S1; 0.10 mm; M 7-12; S1p46	Retain related service value Not the active benchmark0.05 mm limit.
S17: MLS finish recommendation	50 or finer RA; page unit unspecified; service corpus; not an implemented limit	public source-supported; S3; 50 RA or finer; Surface Finishing	Retain source wording without guessing the dimensional unit The benchmark0.5 um value is a separate authored state.
G12: Catalog bore	75.5 mm; service corpus; not selected stock	public source-supported; S6; 75.5 mm; C4251-030 product fields	Retain historical catalog separately from selected stock B01 selected75 mm overrides this related listing for task decisions.
G13: Catalog layer count	3 layers; service corpus; related product	public source-supported; S6; 3; C4251-030 product fields	Product description, not proof of selected stock material response
G14: Catalog orientation field	None catalog text; service corpus; related product	public source-supported; S6; None; C4251-030 product fields	Preserve literal product metadata without inferring reversibility Does not override benchmark up orientation.

Appendix C. Engineering specifications — S18 to J04

Table 25. Appendix C. Engineering specifications — S18 to J04

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
S18: Flatness sampling protocol	head:edges and three center directions;block:edges and across center procedure; service corpus; abstract measurement	public source-supported; M|S1; head:three center directions;block:edges and across center; M 6-45 head; M 7-12 block	Public service protocol; runtime returns a scalar stand-in A scalar warpage value is not an executed straightedge sampling pattern.
J01: Gasket recovery fraction	0.1 fraction; supplemental estimate	benchmark policy choice	Explicit inherited recovery estimate; not selected material calibration No checkpoint or technician-time authority.
J02: New-gasket model history	fresh zero-force initialization; no carried arrays state; supplemental estimate	benchmark policy choice	Explicit installation-epoch initialization Old gasket history is not relabeled as new gasket history.
J03: Bolt-extension interpretation	supplied stretch_mm treated as permanent set model assumption; supplemental estimate	benchmark policy choice	Explicit software transfer from environment truth Not inferred from torque labels or nominal free length.
J04: Gasket thickness transfer	effective uniform-layer strain scaling model assumption; supplemental estimate	benchmark policy choice	Transfer inherited contact law using supplied thickness and unchanged support frame No selected multilayer material curve; passage layout unknown.

Appendix C. Engineering specifications — J05 to D01

Table 26. Appendix C. Engineering specifications — J05 to D01

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
J05: Model reference thickness	0.762 mm; supplemental estimate	benchmark policy choice	Reference for the declared effective-strain transfer Nominal compressed catalog dimension, not measured installed height.
J06: Torque nut factor	0.2 dimensionless; supplemental estimate	benchmark policy choice	Inherited authored torque-to-force coefficient Not measured selected thread/bearing friction.
J07: Torque-law nominal diameter	10 mm; supplemental estimate	benchmark policy choice	Inherited torque relation T=K d F Coupled sequential equilibrium; not ten independent final-target conversions.
J08: Equilibrium residual tolerance	1e-06 N or N m; numerical implementation	benchmark policy choice	Numerical convergence criterion Not physical measurement accuracy.
D01: Disengaged-fastener threshold	1e-09 turns; numerical implementation	benchmark policy choice	Floating-point disengagement comparison Not a measured thread-engagement uncertainty.

Appendix C. Engineering specifications — D02 to G16

Table 27. Appendix C. Engineering specifications — D02 to G16

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
D02: Numeric comparison tolerance	1e-09 same unit as compared values; numerical implementation	benchmark policy choice	Deterministic equality comparison Not instrument accuracy or service tolerance.
S19: Machining eligibility	True boolean; active benchmark	benchmark policy choice	Configured service availability for the benchmark Actual head height, cam-holder clearance and remaining stock are not established.
S20: Accepted head recovery routes	machine_head or replace operation; active benchmark	benchmark policy choice	Two allowed ways to resolve V3
G15: Stock identifiers	HG-075-0762-MLS;HG-078-1200;HEAD-SERVICE;BELT-SERVICE identifiers; active benchmark	benchmark policy choice	Authored distinguishable supply identities; preserve physical replacement identity Names are benchmark stock codes, not verified supplier part numbers.
G16: Extra stock by variant	wrong gasket:V4;new head:V3;new belt:V5 availability; active benchmark	benchmark policy choice	Variant-specific offered supplies

Appendix C. Engineering specifications — I01 to D04

Table 28. Appendix C. Engineering specifications — I01 to D04

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
I01: Reported dimensional units	flatness:m;roughness:um;bolt extension/free length:mm;gasket bore/thickness:mm channel units; active benchmark	benchmark policy choice	Explicit deterministic observation interface Unit conversion in this table does not add measurement resolution.
I02: Instrument labels	flatness;roughness;length_comparator;length_caliper;caliper labels; active benchmark	benchmark policy choice	Describe the requested observation channel A label does not certify a real instrument or a multi-point protocol.
I03: Unprovided bolt free length	null / unknown mm; active benchmark	benchmark policy choice	Preserve missing free-length observation instead of deriving it from torque or nominal length Nominal140 mm part length is not an observed free length.
D03: Root policy metadata	public_observation=technician-v2;qualification=software-fixture;flatness_limit_source_ids=[B1] metadata; active benchmark	benchmark policy choice	Observation exposure and policy provenance; numeric root keys are listed separately Metadata is not additional physical calibration.
D04: Complete delivered removal-rule geometry	source-versioned body poses, routes, precedence and release/restoration rules bound set; complete source-bound machine census, not a sampled service value	benchmark policy choice	Finite CAD-derived obstruction model and declared procedural rules; complete pointer census retained in the audit Grouped geometric arrays are not workshop limits; no whole-engine contact or trajectory certificate.

Appendix C. Engineering specifications — J09 to J12

Table 29. Appendix C. Engineering specifications — J09 to J12

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
J09: Event-count maximum	2048 events; numerical implementation	benchmark policy choice	Bounded deterministic solver workload
J10: Newton iteration maximum	100 iterations; numerical implementation	benchmark policy choice	Bounded solve workload
J11: Line-search step maximum	40 steps; numerical implementation	benchmark policy choice	Bounded numerical line search
S21: Accepted method names	plastic_scraper;solvent;abrasive_disc;thread_brush enum; active benchmark	benchmark policy choice; S4; plastic low-angle preparation; avoid abrasive damage; Surface preparation guidance	Software action domain includes good and adverse actions Acceptance by the interface does not make every method correct; removed fasteners require thread_brush.
J12: Loading-mode name	native-calibrated enum; supplemental estimate	benchmark policy choice	Select inherited authored component contact law The name refers to component-model calibration, not measured selected stock or factory qualification.

Appendix C. Engineering specifications — J13 to C11

Table 30. Appendix C. Engineering specifications — J13 to C11

Specification	Benchmark value and scope	Basis and public-source value	Selection and disagreement or limit
J13: Environment to model position map	hb01..hb10 to bolt_01..bolt_10 by matching ordinal ten explicit bindings; supplemental estimate	benchmark policy choice	Declared correspondence of retained source body identities Replacement physical identities remain distinct from fixed service positions.
H24: Accepted CP04 loosening sequences	10,9,8,7,6,3,4,5,2,1 OR 10,9,8,7,6,5,4,3,2,1 position sequence; current signed benchmark acceptance	benchmark policy choice; M|S1; M printed diagram and S1 exact reverse prose differ; M 6-30 mapped through6-54; S1 printed page16	Amendment 1 accepts either documented sequence because public instructions disagree Source amendment and runtime implementation are separate epochs; old recorded commands are unchanged.
C11: Accepted CP12 cover-gasket condition	new OR original inspected and refitted with specified sealant criterion; current signed benchmark acceptance	benchmark policy choice; M|S1; Inspect gasket condition; local oil and sealant treatment; M 6-55 step12 and application note; S1 printed page17	Amendment 1 withdraws mandatory new cover gasket while retaining mandatory new head gasket Runtime/scorer adoption must be checked at its new source epoch; no historical run is relabeled.

Appendix D. Agent details and harness disclosure

The collector must restrict the agent to the environment MCP interface and record the full prompt, tool schema, requested model, returned model and execution configuration. Session state and credentials must be isolated from unrelated work. The public report does not disclose account identifiers.

The recorded historical collector includes wrapper and process-recovery evidence. A successful owned-process cleanup test does not prove current model identity, account isolation or production lifecycle behavior. These require tests against the exact collector used for study runs.

Record model calls, tool attempts, successful tool actions, prompt and completion tokens, cache accounting if available, wall-clock start and end, retry events and termination reason. If a field cannot be obtained, report it as unavailable and explain the measurement boundary. Do not infer model calls from tool calls or price from subscription access.

Keep configuration and prompt versions alongside every attempt. Retries must retain their prior records. A failure to execute the apparatus is distinct from an agent choosing an incorrect action. Neither a cleanup failure nor a partial response should erase the evidence already collected.

The agent record should distinguish actions, observations and middleware behavior. Document any automatic retries, preprocessing, state recovery or submission logic, so readers can tell which decisions the model made. Store model-call counts separately from tool-call counts, and preserve prompt and completion tokens, dated prices and elapsed time when those measurements are available. Missing telemetry must remain unavailable rather than being inferred from action counts [5] [6].

GPT-6-Luna's finish-action response exceeded the collector's message size limit and was not captured; the game state at its finish action is reported.

Appendix E. Evaluation protocol

This condensed protocol governs the proposed comparison. The complete historical form and a field-by-field amendment mapping accompany the report. The present design replaces earlier survival endpoints, factorial settings and judge-based scoring; historical records are retained without making them current requirements.

Study design and approval The dated study design governs the repair benchmark. Record the approved questions, collection scope and amendments before examining outcomes. Collection beyond the exploratory allowance requires the owner's approval. Preserve excluded and superseded records, record changes and document the authority for publication.

Task definitions and versions Fix the task variants, checkpoint definitions, weights, prompts and evaluator versions before agent collection. Screen each task against the correct reference procedure and review any ambiguous interpretation. Retain earlier versions separately and record the reason for every later change.

Collection environment and validation Describe the collection environment, permitted tools, launch method, configuration and isolation checks. Specify validation frequency, scope and method before collection. Complete the applicable preflight and retain validation evidence for each batch. Tool-interface tests, operating-system isolation and provider qualification are separate checks; earlier fixtures do not establish the selected collection environment.

Agent settings and telemetry Record requested and returned model identities, collection surface, prompts, resource limits and supported decoding settings. Preserve commands, diagnostics and revisions. Steps counts model calls; tool calls and interface turns are separate measures. Unavailable identity or usage fields remain unknown.

Execution order and resources State funding, execution order, concurrency and retry scheduling. Give each attempt a unique identity, prevent duplicate starts and record scheduling changes after errors. Historical defaults are not the new approved collection plan. Report subscription use separately from token-based API-equivalent cost.

Variants and sample allocation Specify the model-by-variant allocation, exploratory allowance, repeats and completion target before collection. The proposed variants remain draft until their definitions and implementations are validated. Retain charged attempts in the accounting, and obtain approval before expanding the full collection matrix.

Interrupted attempts and retries Retain every attempt, including interruptions, provider failures and refusals. Classify model outcomes separately from apparatus failures. Fix inclusion, retry and response-selection rules, deduplicate responses and reconcile batch totals. Report observed checkpoint evidence and named unknowns for incomplete attempts; do not silently remove them from the report or assign missing observations a zero score.

Reference procedure and extraction Fix reference inputs and extraction rules, and use unique run identifiers. Validate checkpoint decisions against the correct scripted procedure and targeted defective jobs, with independently reviewed final state and action histories. Record timeouts, completion evidence and evaluator revisions. The correct reference must pass every applicable checkpoint; this requirement is separate from an observed validation result.

Ambiguity and refusal Distinguish an agent refusal, a clarification request, an unsupported task and a collection failure. State the classification rule and preserve the original response. An unresolved apparatus specification is not automatically an agent mistake. Omitting a required agent action under a valid fixed reference can fail its checkpoint.

Scoring and uncertainty Use the template's full and partial completion measures. State the Success denominator and Wilson interval, and the bootstrap sampling unit for Score and technician time. Fix variant weighting, incomplete-attempt treatment and resampling settings before collection. Preserve dependence among checkpoints and bolts within a run. Earlier survival and factorial plans are superseded; exploratory checks do not establish general model rankings.

Reporting and deviations Report counts through collection and analysis, separate exploratory and later cohorts, and explain deviations. Show model calls, tool calls, tokens and wall time separately. Report productive technician time, typed corrective penalties and the correct reference time, preserving the distinction between cited durations and estimates. Agent results remain pending collection; apparatus records are reported separately.

Evaluator review This application uses deterministic checkpoint evaluators. Independently review their decisions against final state and chronological records under a stated manual-review plan. The earlier language-model judge threshold and independence settings do not apply to this task. The manual-review sample and selection rule must be specified rather than inferred from the historical form.

Appendix E. Current exploratory collection status

The protocol summary above preserves the reviewed precollection design. Exploratory repair observations are now reported separately in the results sections. The larger comparative collection and confirmatory analysis remain pending. These observations do not amend the historical form, remove review requirements or establish physical repair safety.

Appendix F. Exploratory resources

One exploratory attempt was observed for each requested model on the standard repair task. The model names identify the model; served model identity was not attested. The collector does not supply model-call Steps or a model-specific API price basis, so those cells remain unavailable. A stopped repair remains in the run count.

Table 31. Appendix F. Exploratory resources

Model requested	Tools attempted	Productive s	Penalty s	Charged s	Mistake events
GPT-6.1-Sol	663	13280	12012	25292	245
GPT-6-Astra	528	16190	1404	17594	30
GPT-6-Luna	807	15230	14970	30200	309

Table 32. Appendix F. Exploratory resources

Model requested	Agent wall s	Input tokens	Output tokens	Cache-read tokens
GPT-6.1-Sol	1079.565809254	3877499	22686	3767168
GPT-6-Astra	555.097799603	2301314	8750	2187520
GPT-6-Luna	Unavailable	13048986	62622	12776448

Productive and penalty charges are additive. Mistake events count affected actions; typed labels can overlap. Cache-read tokens are already included in input tokens and must not be added again. Agent wall time excludes setup; tool attempts are not model-call Steps. Recorded telemetry and unavailable observations retain separate meanings.

Luna token counters are the latest observed prefix, with final-turn coverage unavailable. They are displayed separately from complete-turn resource measurements and do not supply a complete consumption estimate. Its agent wall time remains unavailable. The accepted finish endpoint preserves the original capture refusal and pending response record; it does not convert that record into a completed model turn.

For Sol, the independent scorer recorded 25262 charged seconds, whereas the used in-game classification records 25292. The documented difference is a cover-gasket refit classified as productive work by the independent scorer and as a mistake penalty in the game. The original records are retained. The table uses the in-game policy; primary checkpoint points agree.

The independent and game scorers use different secondary checklist item sets and retain unknowns. Agreement on primary points and shared bolt fields does not establish full service coverage or external contract identity. Astra and Luna shared charge entries agree; Sol retains the stated limitation. Luna also retains an overlapping mistake-label difference between scorers. These model attempts remain separate from the original-rule scripted references and the earlier incomplete diagnostic jobs.

Appendix F. Exploratory checkpoint outcomes

Each column is one repair attempt. Outcomes join checkpoint identity, weight and pass status between the game and independent reports. The full requirements appear in Appendix C.

Table 33. Appendix F. Exploratory checkpoint outcomes

Checkpoint	Weight	GPT-6.1-Sol	GPT-6-Astra	GPT-6-Luna
CP01	1	Pass	Pass	Pass
CP02	2	Fail	Fail	Fail
CP03	1	Pass	Pass	Pass
CP04	1	Fail	Fail	Pass
CP05	1	Fail	Pass	Fail
CP06	1	Pass	Pass	Pass
CP07	1	Fail	Pass	Pass
CP08	2	Fail	Pass	Pass
CP09	2	Fail	Pass	Pass
CP10	2	Fail	Pass	Pass
CP11	2	Fail	Pass	Pass
CP12	2	Fail	Pass	Fail

Appendix F. Exploratory head-bolt checks

Table 34. Appendix F. Exploratory head-bolt checks

Head-bolt position	GPT-6.1-Sol	GPT-6-Astra	GPT-6-Luna
hb01	0/6	6/6	6/6
hb02	0/6	6/6	6/6
hb03	0/6	6/6	6/6
hb04	0/6	6/6	6/6
hb05	0/6	6/6	6/6
hb06	0/6	6/6	6/6
hb07	0/6	6/6	6/6
hb08	0/6	6/6	6/6
hb09	0/6	6/6	6/6
hb10	0/6	6/6	6/6

Cells give raw criteria passed out of criteria checked. The raw record keeps bolt condition, initial and final torque values, each stage sequence and stage order separate. The study collector’s reduced mapping combines the two sequence checks; its denominator is different. Sol’s removed bolts are an unfinished observation; final accuracy remains unavailable. Correct final bolt checks do not establish correct disassembly, loosening or restoration.

Appendix F. Exploratory mistake labels

Table 35. Appendix F. Exploratory mistake labels

Mistake type	GPT-6.1-Sol	GPT-6-Astra	GPT-6-Luna
blocked removal	3	1	7
invalid action	28	14	110
missing attachment restore	None recorded	None recorded	1
redo work	213	15	191
wrong gasket	2	None recorded	1

An action may carry several labels. The unique event count in the resource table therefore need not equal the sum of these categories. Corrective penalties remain charged even if the agent subsequently corrects the state.

Appendix F. Results machinery and scripted checks

The results software reads saved report data and generates aggregate tables, per-variant and per-checkpoint summaries, per-bolt records, typed mistakes and figures. It does not execute the environment or an agent. The cases below test that reporting path; they are historical scripted procedures, not language-model outcomes.

Table 36. Appendix F. Results machinery and scripted checks

Saved scripted case	Points	Productive seconds	Penalty seconds	Charged seconds
Historical baseline (incomplete)	12	5860	0	5860
wrong torque sequence	8	5860	600	6460
under torque	10	5860	120	5980
skipped final stage	10	5740	12	5752
wrong gasket	10	5860	30	5890
reused stretched bolt	9	5830	30	5860
wrong timing	12	5860	120	5980

The case name correct is inherited from its producer; the recorded draft checkpoints show that the baseline is incomplete. The reports lack model, agent-arm and current-variant identities. They cannot establish the revised correct reference or populate a model-by-variant results table. The timing fault also shares an already unmet timing checkpoint with the incomplete baseline, so equal aggregate scores do not demonstrate evaluator sensitivity to that fault.

Charges are analyst task estimates, not measured technician time or a published labor-time standard. Model-call Steps, token counts, API-equivalent Cost and agent wall time are unavailable in these reports. The companion tables preserve those missing values and their reasons; tool attempts remain a separate field. Scripted case means have no inferential intervals and do not populate common-failure findings.

For later model runs, every row must retain the collector identity, task and evaluator versions, original report and telemetry provenance. The reducer keeps interrupted and unsuccessful attempts, checks weighted checkpoint arithmetic and labor-ledger totals, and preserves unknown component observations. Complete means are withheld when a component is missing; any available-case mean has its own denominator. The prepared analysis resamples whole runs within declared variants, never individual bolts.

Appendix F. Original-rule scripted references

These saved procedures complete the original signed checkpoint rules in the environment scorer and the independent scorer. They are scripted references, not model attempts. Each uses the earlier disassembly policy and requires both gaskets to be new; the later allowance for an inspected cover gasket and the grouped rocker route were not applied retroactively.

Table 37. Appendix F. Original-rule scripted references

Variant	Environment points	Independent points	Tool actions	Productive seconds	Penalty seconds	Charged seconds
V1	18/18	18/18	450	14050	0	14050
V2	18/18	18/18	451	14080	0	14080
V3	18/18	18/18	452	15310	0	15310
V4	18/18	18/18	452	14140	0	14140
V5	18/18	18/18	451	14080	0	14080

Both scorers report full completion 1 and partial score 1.0 for each listed reference. Tool actions are not model-call Steps. Productive and penalty seconds are analyst charges, not elapsed mechanic time. The machining route carries a larger charge; the separately recorded replacement-head route is a different procedure and is not pooled into this table.

Checkpoint outcomes and shared bolt, time and mistake values agree. This does not make the secondary checklists equivalent: the standard case has 1674 environment items and 2799 independent items, with unknown condition observations retained. Evidence descriptions also differ. Full service coverage and modeled checklist completion are not established. The comparison leaves external contract authentication as a separate requirement. These scripts record zero model and provider calls and zero reported API-equivalent cost. A separately computed token-priced Cost remains unavailable without dated rates. Agent wall time is unrecorded.

The older incomplete scripted cases above retain their original records. Neither cohort supplies agent performance or empirical model-failure rates. Runs under the amended rules require their own runtime and evaluator evidence.

Appendix G. Limitations

One run per model; descriptive, not comparative. Technician-time charges are estimates requiring validation against published labour-time guides. Game performance is not physical repair validation.

The model observations are exploratory attempts on a standard configured task. Requested model identifiers are known, but served identities and model-call Steps are unavailable.

Geometric visibility, software action checks and finite mechanical predictions establish different aspects of the apparatus; they do not certify physical repair. Factory service applicability, physical machining and fit, full service coverage, external contract identity and professional labor-time sources remain qualified. Earlier incomplete diagnostics and complete original-rule references stay separate from amended-rule model attempts. The larger comparative study remains pending.

The benchmark covers a prepared engine and a narrow repair procedure. It omits vehicle-level access, physical dexterity, workshop hazards and many inspection cues. Success in this game cannot certify a physical repair or authorize deployment on real equipment.

Geometry is detailed but service access, component mapping and legal transitions are incomplete in the reviewed snapshot. Complete original-rule software references are available for the configured abnormal conditions; equivalent coverage under the amended rules remains to be demonstrated. Service thresholds and orientation information must be resolved rather than invented.

Mechanical models use generic material properties and finite comparisons. They support selected state estimates but are not selected-part durability validation. No operating-engine endpoint contributes to scoring.

Technician charges are a domain cost estimate rather than measured workshop performance. The historical action table lacks the required professional time sources and correction schedule; later scripted diagnostic charges remain analyst estimates. API-equivalent cost may be unavailable for a returned model; that limitation must not be concealed by a subscription-price proxy.

A small set of related variants cannot establish broad repair competence or occupational automation. Repeated runs, within-repair bolt observations and shared task structure create dependence. The final analysis must state its population and sampling limits; exploratory checks must not be presented as a confirmatory study [7] [8].

Data and code availability: the report is accompanied by a source index, numerical claim table, evaluation specification, protocol amendment mapping and configuration records. Source identifiers and complete content hashes are kept in these companions so that each substantive claim can be inspected without turning the paper into a process log.

Appendix H. Data and artifacts

Table 38. Appendix H. Data and artifacts

Table or figure	Record	File
Figure 1	Original illustration	data/received/workflow-diagram.png
Table 1	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 2	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 3	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 4	Complete source and quantity ledger	manuscript/claim-to-data.csv
Figure 2	Original illustration	data/received/oil-render.png
Figure 3	Original illustration	data/received/cad-iso.png
Figure 4	Original illustration	data/received/joint-laws-figure.png

Appendix H. Data and artifacts continued

Table 39. Appendix H. Data and artifacts continued

Table or figure	Record	File
Table 5	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 6	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 7	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 8	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 9	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 10	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 11	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 12	Complete source and quantity ledger	manuscript/claim-to-data.csv

Appendix H. Data and artifacts continued

Table 40. Appendix H. Data and artifacts continued

Table or figure	Record	File
Table 13	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 14	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 15	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 16	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 17	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 18	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 19	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 20	Complete source and quantity ledger	manuscript/claim-to-data.csv

Appendix H. Data and artifacts continued

Table 41. Appendix H. Data and artifacts continued

Table or figure	Record	File
Table 21	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 22	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 23	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 24	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 25	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 26	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 27	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 28	Complete source and quantity ledger	manuscript/claim-to-data.csv

Appendix H. Data and artifacts continued

Table 42. Appendix H. Data and artifacts continued

Table or figure	Record	File
Table 29	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 30	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 31	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 32	Complete source and quantity ledger	manuscript/claim-to-data.csv
Table 33	Paired repair reports	build/results/real/results.json
Table 34	Paired repair reports	build/results/real/results.json
Table 35	Paired repair reports	build/results/real/results.json
Table 36	Complete source and quantity ledger	manuscript/claim-to-data.csv

Appendix H. Data and artifacts continued

Table 43. Appendix H. Data and artifacts continued

Table or figure	Record	File
Table 37	Complete source and quantity ledger	manuscript/claim-to-data.csv
Common failures	Approved trajectory analysis	data/received/approved-common-failures.md
Common failures	Action witnesses	data/received/approved-common-failure-sequences.json
Scripted reference	Amended-rule completed standard-task trace	data/received/editorial-amended-reference.json
Scorer corrections	Current per-action charge and checkpoint agreement	data/received/editorial-current-scorer-check.json
Task design and scoring	Benchmark design	data/received/study-design.md
Service information	Specifications and source corpus	data/received/service.json
Exploratory reports	All paired records and public identity/capture inputs	data/results-inputs.json

Appendix H. Data and artifacts continued

Table 44. Appendix H. Data and artifacts continued

Table or figure	Record	File
Scripted references	Original-rule full reference pairs	data/reference-inputs.json
Table 38	Data and artifacts source map	data/paper.json
Table 39	Data and artifacts source map	data/paper.json
Table 40	Data and artifacts source map	data/paper.json
Table 41	Data and artifacts source map	data/paper.json
Table 42	Data and artifacts source map	data/paper.json
Table 43	Data and artifacts source map	data/paper.json
Table 44	Data and artifacts source map	data/paper.json